

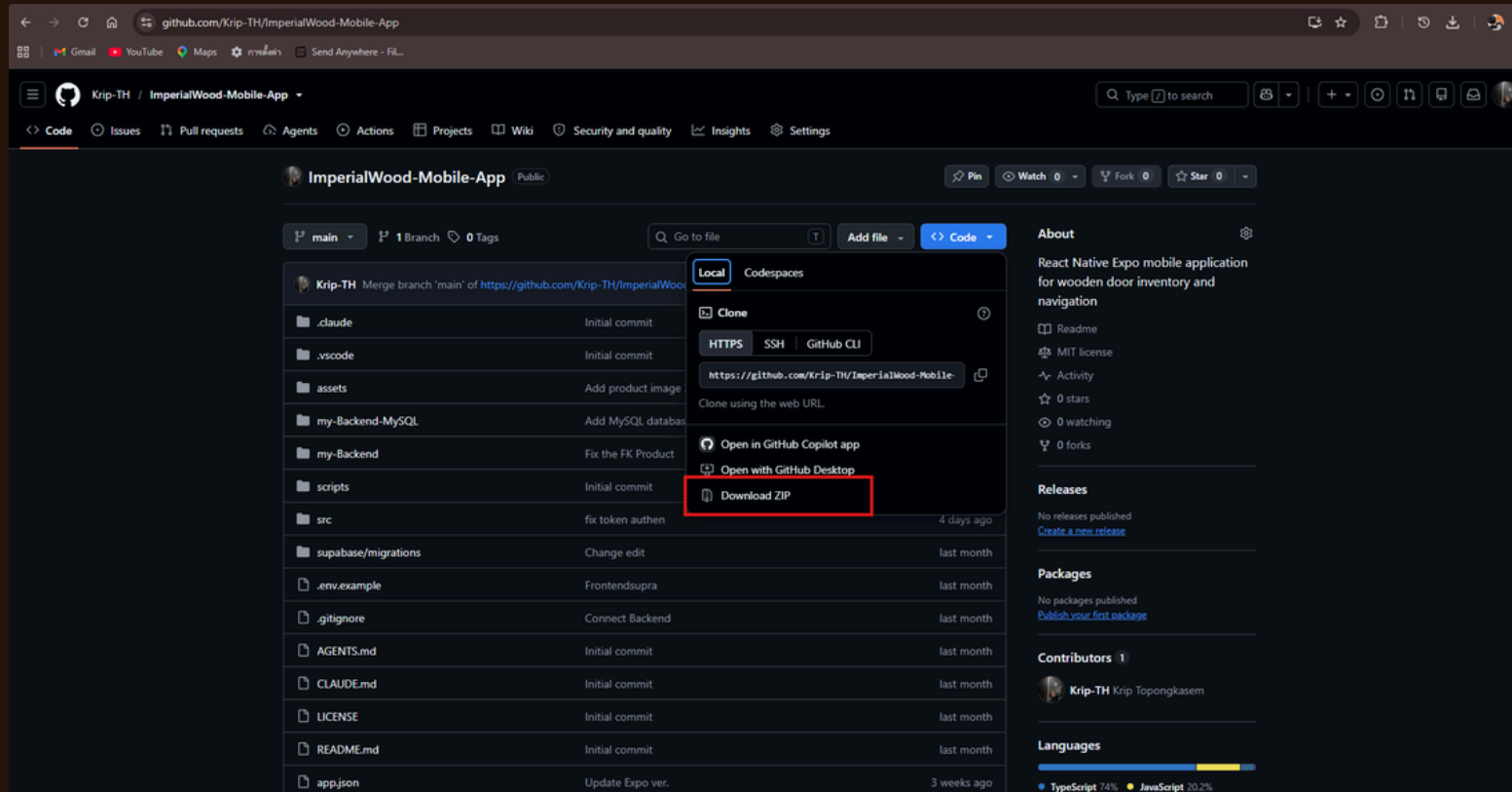
USER MANUAL



Krip Topongkasem

@imperialwood

1. HOW TO DOWNLOAD THE PROJECT FROM GITHUB



1. Open the ImperialWood-Mobile-App repository on GitHub.
2. Click the Code button.
3. Choose a download method :
 - Download ZIP: Click Download ZIP to download the entire project, then extract the ZIP file.
 - Git Clone: Copy the repository URL and run the following command:

```
</> Bash  
  
git clone <repository-url>
```

4. Open the downloaded ImperialWood-Mobile-App folder in Visual Studio Code.



1.2 Required Software, Dependencies, and Configurations

DEVELOPMENT & EXECUTION TOOLS



Node.js & npm

Install and run the frontend and backend components.



Visual Studio Code

Primary environment for editing and managing project source code.



Expo & Android Studio

Tools to run, test, and emulate the React Native application.

DATA & VERSION CONTROL



MySQL & phpMyAdmin

Systems to store, manage, and import the application's database.



Git

Essential for cloning the project and managing versioning.



GitHub (via Git)

Used to clone and manage the project repositories via Git.

FRONTEND

Frontend Dependencies

A visual overview of the primary technologies and libraries used to build the ImperialWood mobile application interface. The frontend is developed using React Native with Expo, identifying dependencies for UI, navigation, local storage, and file handling.



React Native

Used to build the mobile application user interface.



Expo

Used to develop and run the mobile application.



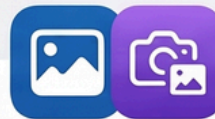
Expo Router

Manages navigation between various application screens.



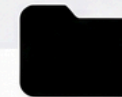
AsyncStorage

Provides a system to store local data on the device.



Expo Image & Image Picker

Used to select, display, and optimize images from the device.



Expo File System

Provides access to manage files stored directly on the device.

All required frontend dependencies can be installed automatically by running :

```
</> Bash
```

```
npm install
```



BACKEND: ImperialWood Core Technologies

A high-level overview of the ImperialWood backend ecosystem, illustrating how Node.js and Express.js power the application logic, communicating with a MySQL database managed via phpMyAdmin.

Node.js



Node.js

The backend runtime environment driving the application logic.



CORS

Enables frontend-backend communication

Express.js

express

The REST API framework driving the application logic.



dotenv

Manages environment variables

MySQL Database



MySQL

The primary system used for structured data storage and persistence.



MySQL2

Connects Node.js to the database

phpMyAdmin



The essential tool used for managing and interacting with the MySQL database.



bcryptjs

Secures user passwords

Install the backend dependencies with:

```
</> Bash
```

```
cd my-Backend-MySQL  
npm install
```



BACKEND CONFIGURATION: ImperialWood Setup Guide

ENVIRONMENT CONFIGURATION



Initialize Local Environment

Rename **.env.example** to **.env** to begin local configuration for the cloud server.



Essential Database Variables

DB_NAME: ip_std6730202025
PORT: 3117

For the backend server connection.



Security Warning: Protect Credentials

Do not upload the **.env** file to GitHub as it contains private credentials.

BACKEND ARCHITECTURE & TECH STACK

Application Request Flow



Mobile App



REST API



Node.js/Express.js



MySQL Database



phpMyAdmin: Management Only

A separate tool used to view and maintain the MySQL database, not the request flow.

Core Technologies

Built with Node.js, Express.js, MySQL, and GitHub for version control.

Example configuration:

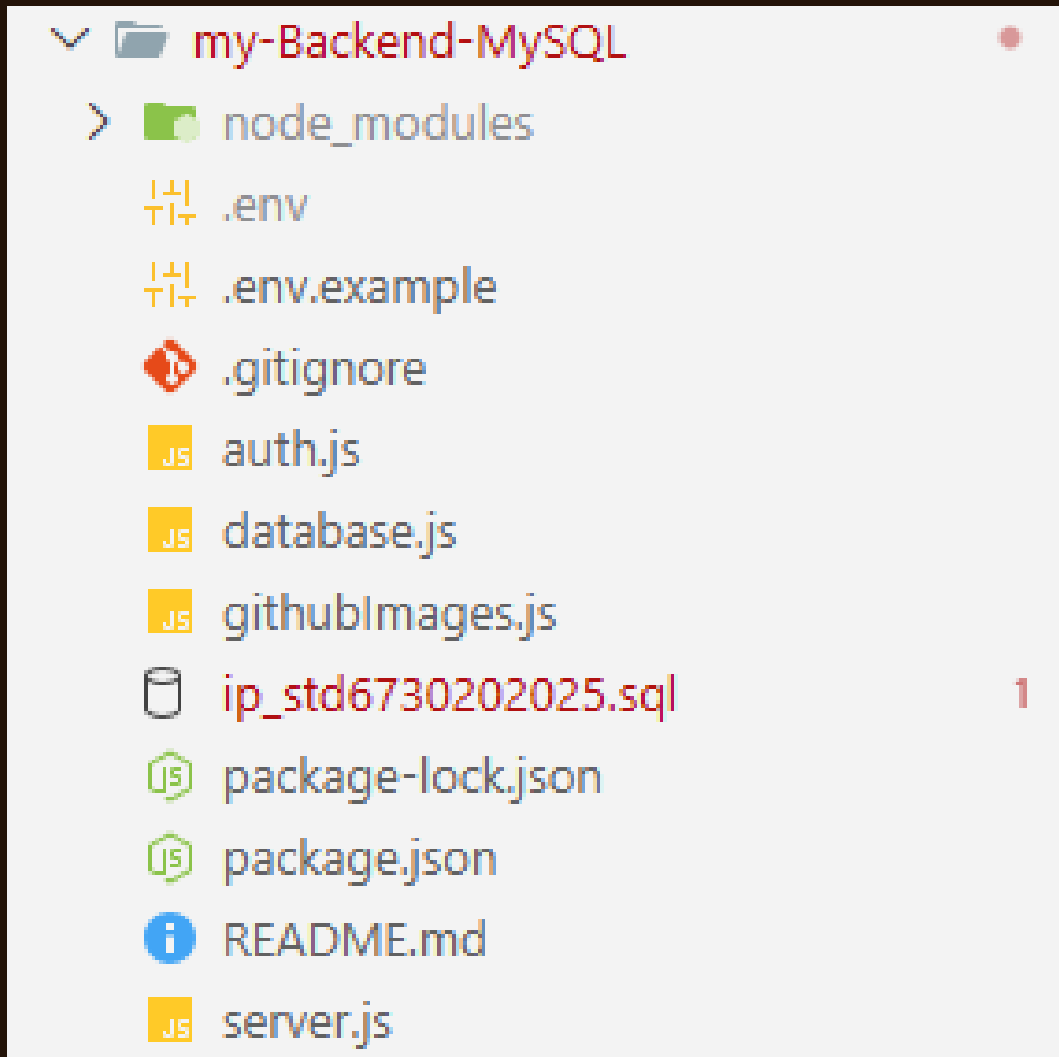
`</> env`

```
DB_HOST=localhost
DB_PORT=3306
DB_NAME=ip_std6730202025
DB_USER=your_mysql_username
DB_PASSWORD=your_mysql_password

PORT=3117

GITHUB_TOKEN=your_github_token
GITHUB_OWNER=Krip-TH
GITHUB_REPO=ImperialWood-Mobile-App
GITHUB_BRANCH=main
```

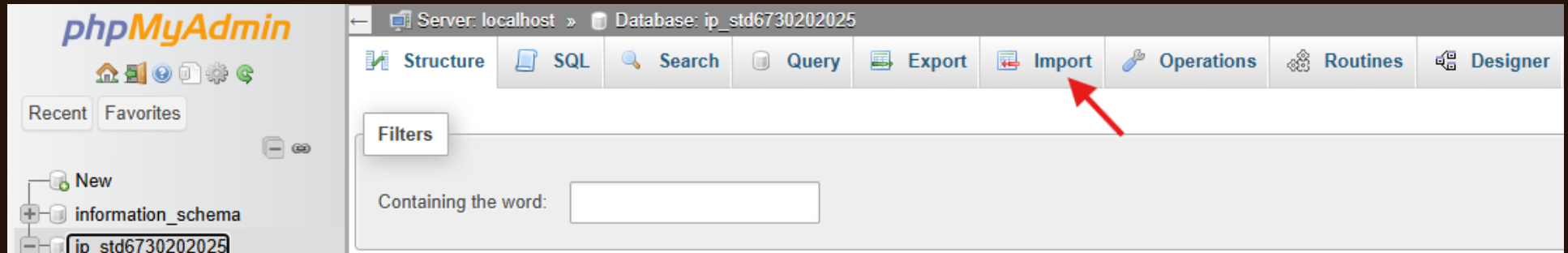
Backend Project Structure in Visual Studio Code



- server.js — Starts the backend server and provides REST API endpoints.
- database.js — Configures the connection to the MySQL database.
- auth.js — Handles user authentication.
- githubImages.js — Handles product image-related functions.
- .env.example — Provides an example of the required environment configuration.
- ip_std6730202025.sql — MySQL database export file for the ImperialWood application.
- package.json — Contains backend dependencies and scripts.

The backend folder contains the server, database connection, authentication, configuration, and MySQL export files.

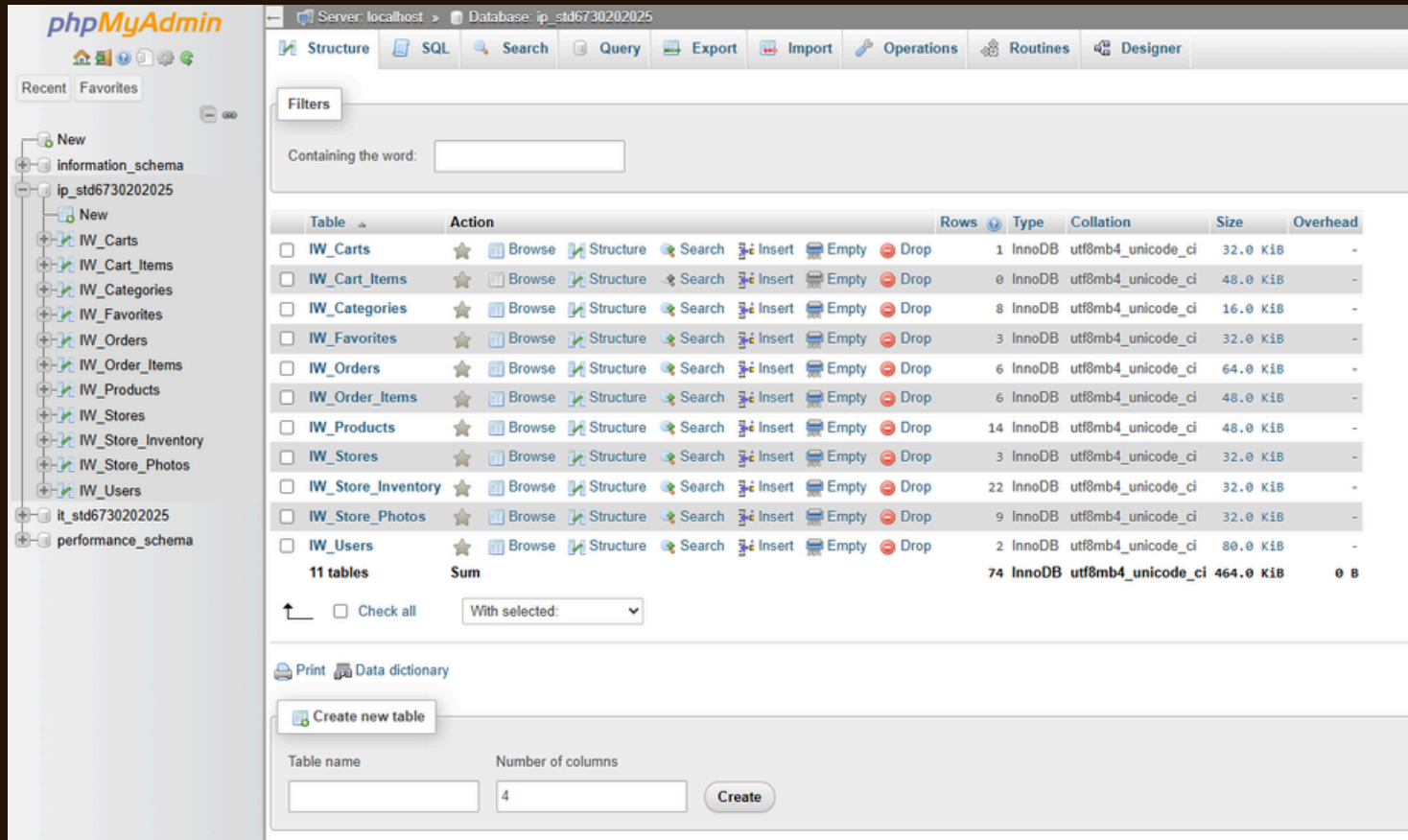
Database Setup — Import MySQL Database



Steps to Import the MySQL Database

1. Open phpMyAdmin.
2. Select the ip_std6730202025 database from the left panel.
3. Click the Import tab.
4. Click Choose File.
5. Select the ip_std6730202025.sql file from the project.
6. Make sure the file format is set to SQL.
7. Click Import to start importing the database.
8. After the import is complete, open the Structure tab to verify that all tables have been created successfully.

Database Import Verification



The screenshot shows the phpMyAdmin interface with the 'Structure' tab selected for the database 'ip_std6730202025'. The left sidebar shows the database hierarchy, including 'information_schema', 'ip_std6730202025', and 'performance_schema'. The main area displays a list of 11 tables in the database, each with a checkbox for selection and a star icon for favorites. The tables are: IW_Carts, IW_Cart_Items, IW_Categories, IW_Favorites, IW_Orders, IW_Order_Items, IW_Products, IW_Stores, IW_Store_Inventory, IW_Store_Photos, and IW_Users. Below the table list, there is a 'Sum' row showing the total size of the database as 464.0 KiB. At the bottom, there is a 'Create new table' button and a form to create a new table with a table name and number of columns (set to 4).

Table	Action	Rows	Type	Collation	Size	Overhead
☐ IW_Carts		1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ IW_Cart_Items		0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ IW_Categories		8	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ IW_Favorites		3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ IW_Orders		6	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ IW_Order_Items		6	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ IW_Products		14	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ IW_Stores		3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ IW_Store_Inventory		22	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ IW_Store_Photos		9	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ IW_Users		2	InnoDB	utf8mb4_unicode_ci	80.0 KiB	-
11 tables	Sum	74	InnoDB	utf8mb4_unicode_ci	464.0 KiB	0 B

After importing the SQL file, the Structure page in phpMyAdmin displays all tables in the ip_std6730202025 database. The database contains 11 tables used to manage users, products, categories, stores, inventory, favorites, shopping carts, and orders.

This confirms that the ImperialWood MySQL database has been imported successfully and is ready for use by the backend application.

Run the Backend on the Cloud Server

Steps to Run the Backend

1. Open the Terminal in Visual Studio Code.
2. Connect to the Cloud Server using SSH:

 Bash



```
ssh -p 2222 std6730202025@119.59.102.161
```

3. Enter the server password when prompted.
4. Navigate to the application workspace:

 Bash



```
cd /app
```

5. Start the Node.js backend server:

 Bash



```
node server.js
```

6. Verify that the terminal displays:

```
Server running on port 3117
```



7. Keep the backend server running while using the ImperialWood application.

Run the Backend on the Cloud Server

Problems 1 Output Debug Console Terminal Ports SQL History Task Monitor

```
PS D:\ImperialWood> ssh -p 2222 std6730202025@119.59.102.161
std6730202025@119.59.102.161's password:
IP course shell: std6730202025
Workspace: /app
Assigned backend port: 3117
Node.js: npm ci then node server.js
FastAPI: fastapi-init then ./run.sh
FastAPI docs: http://119.59.102.161:3117/docs
std6730202025@ip-course:~$ cd /app
std6730202025@ip-course:~$ node server.js
Public backend URL (after successful startup): http://119.59.102.161:3117/
◇ injected env (12) from .env // tip: % suppress logs { quiet: true }
ImperialWood MySQL API listening on port 3117
```

Run the Frontend

Steps to Run the Frontend

1. Open the **ImperialWood-Mobile-App** project in Visual Studio Code.
2. Open a new Terminal in the main project directory.
3. Install the required frontend dependencies:

 **Bash**



```
npm install
```

4. Start the Expo development server:

 **Bash**



```
npx expo start
```

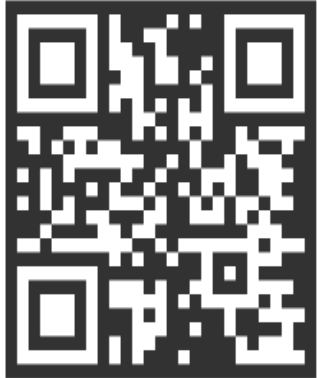
5. Wait for the Expo development server to start.
6. Open the application using an **Android Emulator**.
7. Verify that the ImperialWood application starts successfully and can communicate with the backend server.

Description

The ImperialWood frontend runs using React Native with Expo. After the Expo development server starts, the application can be launched on an Android Emulator for testing. The Cloud Backend must remain active so that the application can access data through the REST API.

Run the Frontend

```
Starting project at D:\ImperialWood
Using src/app as the root directory for Expo Router.
React Compiler enabled
Starting Metro Bundler
```



```
> Scan the QR code above to open in Expo Go.
> Metro: exp://192.168.31.67:8081
> Web: http://localhost:8081
```

```
> Using Expo Go (Press s to switch to development build)
> Press ? | show all commands
```

```
An update for expo is available: 57.0.11 → ~57.0.18
13 other packages may need updating. Run npx expo install --check for details.
Logs for your project will appear below. Press Ctrl+C to exit.
```

```
> Opening on Android...
```

```
> Opening exp://192.168.31.67:8081 on Medium_Phone_2
```

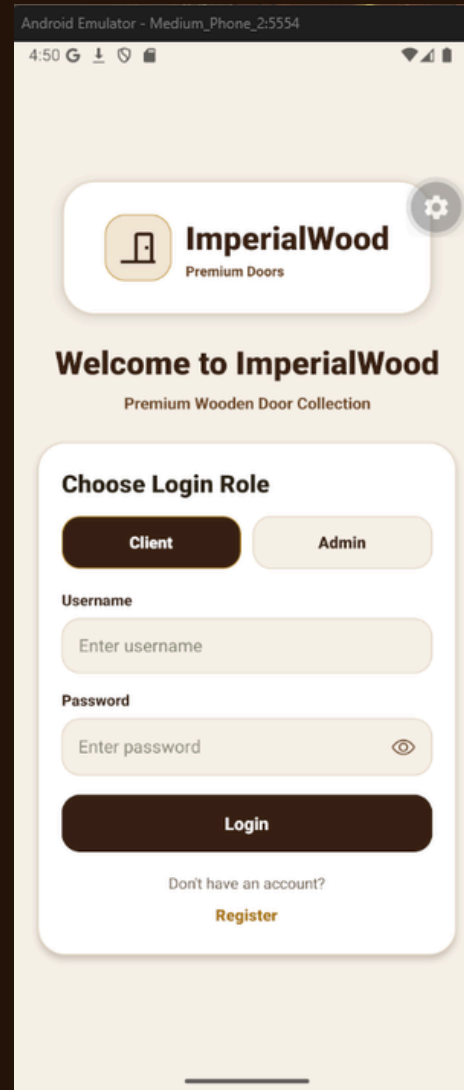
```
✓ Expo Go 57.0.9 is recommended for SDK 57.0.0 (Medium_Phone_2 is using 57.0.3). Learn more: https://docs.expo.dev/develop/tools/#expo-go. Install the recommended Expo Go version? ... no
```

```
> Press ? | show all commands
```

```
Android Bundled 1269ms node_modules\expo-router\entry.js (1862 modules)
```

The ImperialWood frontend is started using the `npx expo start` command. Expo launches the Metro development server and provides a QR code for running the application. The message “Android Bundled” confirms that the application has been successfully built and loaded for the Android environment.

Run the Frontend



Running the Application on Android Emulator

After the frontend is started with Expo, the ImperialWood application can be launched on an Android Emulator. The successful display of the Login screen confirms that the frontend has loaded correctly and is ready for use.

Application Navigation Flow

ImperialWood Application: Navigation & User Flow

Client-Side Experience

The Purchase Journey



Primary Navigation Bar



Utility Views



Login / Client Registration



Feature	Client Access	Admin Access
Cart / Place Order	✓	✗
Add / Edit Products	✗	✓
Store Info / Categories	✓	✓

Admin Management

Product Management Flow



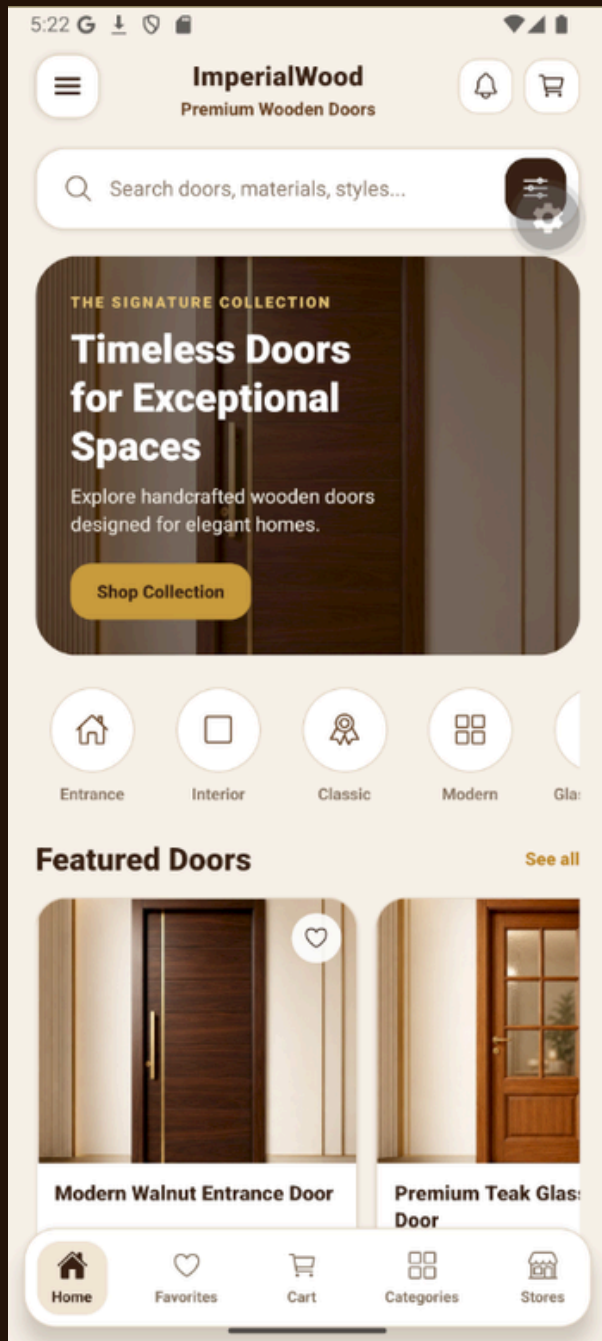
Creation Workflow



Admin Dashboard & Navigation



Client Navigation and Main Features



The Client Home screen provides access to ImperialWood's main shopping features. Users can browse, search, and manage products through the main interface and bottom navigation bar.

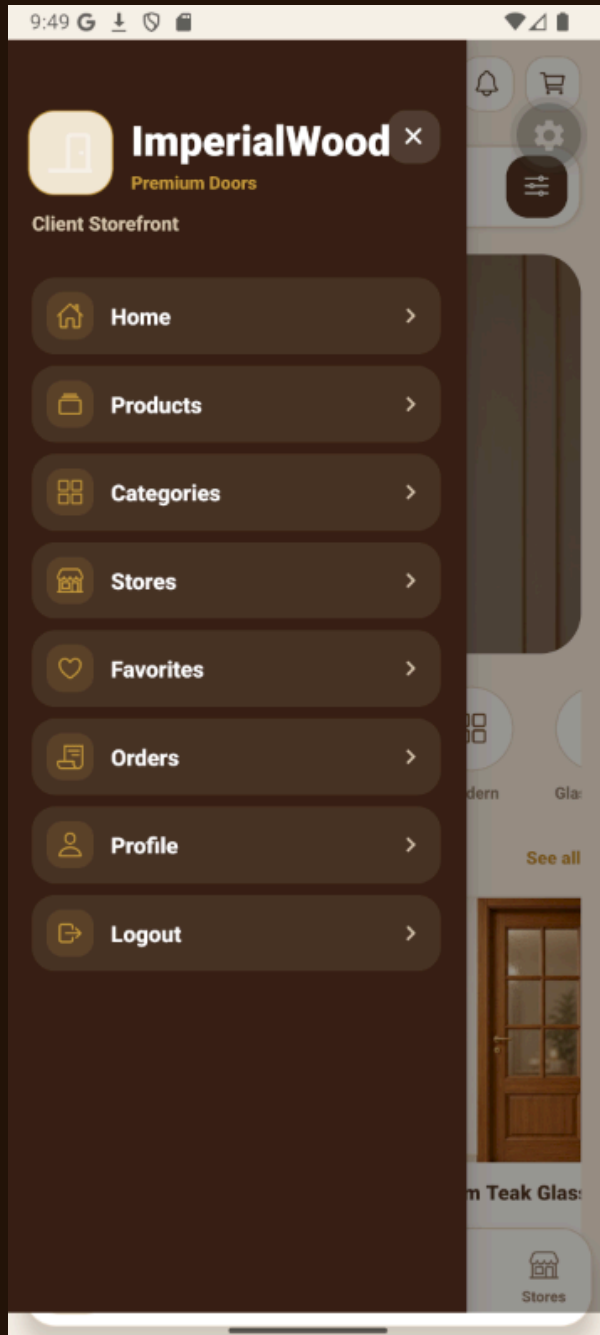
Main Features:

- Search & Filter — Find products and refine the displayed results.
- Product Categories — Browse doors by category or style.
- Featured Doors — View highlighted products directly from the Home screen.
- Favorites — Save preferred products for quick access.
- Cart — View and manage selected products.
- Categories — Browse available product categories.
- Stores — Access ImperialWood store information.
- Menu — Provides access to additional application options.

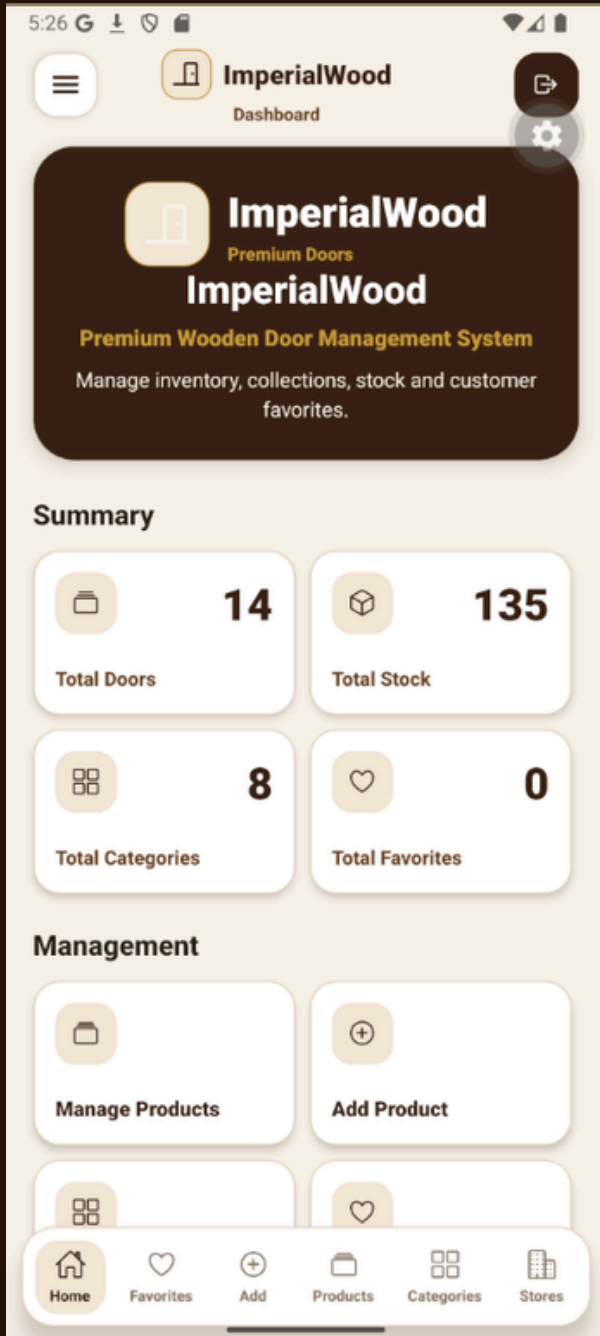
The Client navigation menu provides

Client Navigation Menu

The Client navigation menu provides quick access to the main shopping and account functions of ImperialWood. Clients can navigate between Home, Products, Categories, Stores, Favorites, Orders, and Profile, or select Logout to exit the current session.



Admin Navigation and Main Features



The Admin Dashboard provides an overview of ImperialWood inventory and management functions. Administrators can monitor product and stock information and quickly access product management features through the dashboard and bottom navigation bar.

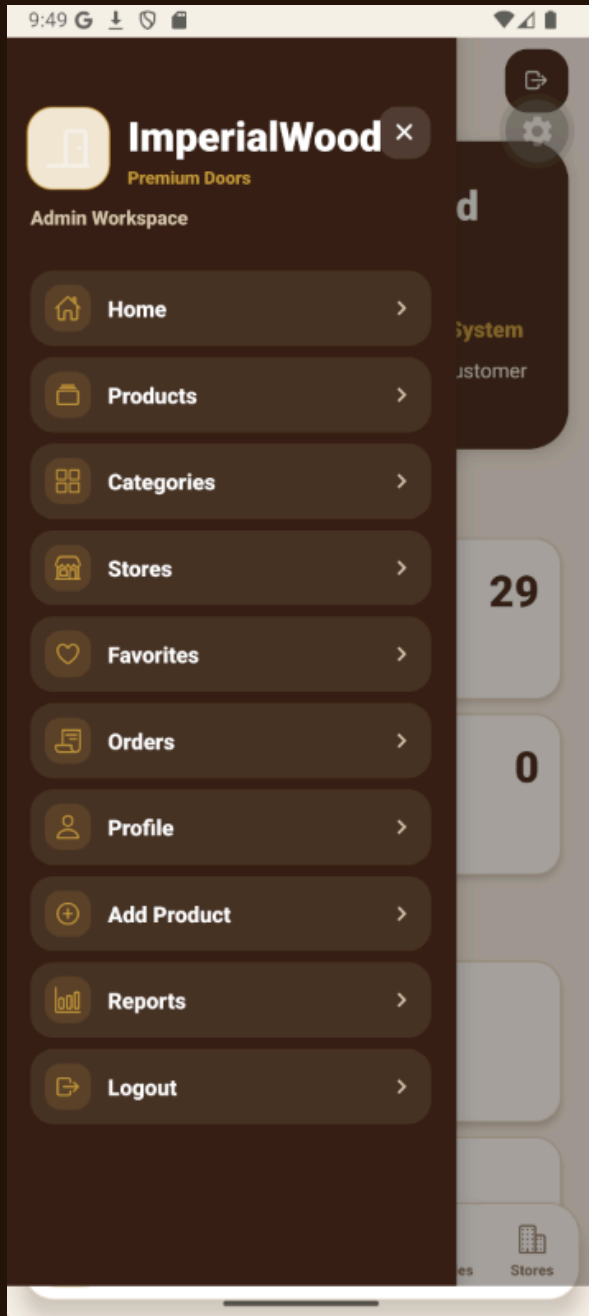
Main Features:

- Dashboard Summary — Displays an overview of total doors, stock, categories, and favorites.
- Manage Products — Allows the administrator to view and manage product information.
- Add Product — Allows the administrator to add new products to the system.
- Favorites — Provides access to saved favorite products.
- Products — Displays the product list for management.
- Categories — Provides access to product categories.
- Stores — Displays ImperialWood store information.
- Menu — Opens additional navigation and administration options.
- Logout — Allows the administrator to securely exit the current session.

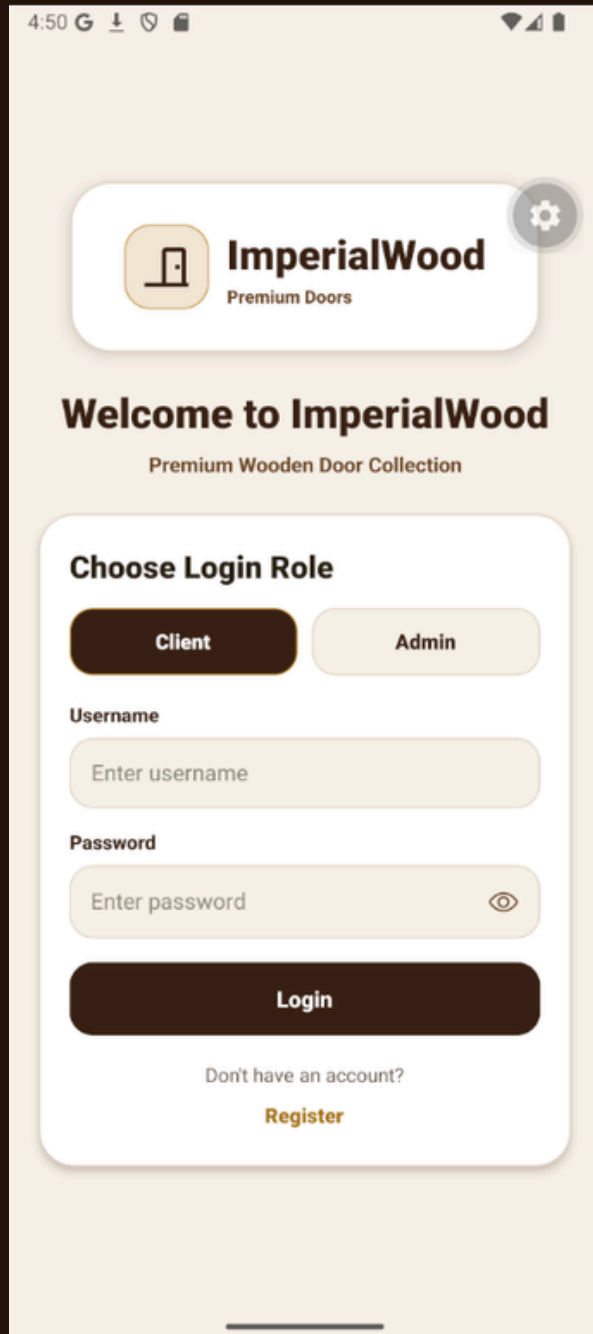
The Admin navigation menu provides

Admin Navigation Menu

The Admin navigation menu provides access to both general application pages and administrative functions. Administrators can access Home, Products, Categories, Stores, Favorites, Orders, Profile, Add Product, and Reports, or select Logout to exit the current session.



Client



Page-by-Page Description

Login Page

The Login page is the main entry point to the ImperialWood application. It allows users to select a role and sign in to access the appropriate Client or Admin interface.

Features & Functions

- Role Selection — Select Client or Admin before signing in.
- Username — Enter the registered username.
- Password — Enter the account password.
- Show/Hide Password — Display or hide the entered password.
- Login — Authenticate the user and open the appropriate interface.
- Register — Allows a new Client user to create an account.

How to Use

Select Role → Enter Username → Enter Password → Tap Login → Access Client Home or Admin Dashboard

10:05 G 4G+ 100% 100%

ImperialWood

Create Customer Account

Join ImperialWood and explore premium wooden doors.

Full Name
Enter your full name

Username
Choose a username

Email
Enter your email

Phone Number
Enter your phone number

Password
Create a password

Confirm Password
Re-enter your password

Create Account

Already have an account?

Page-by-Page Description

Register Page

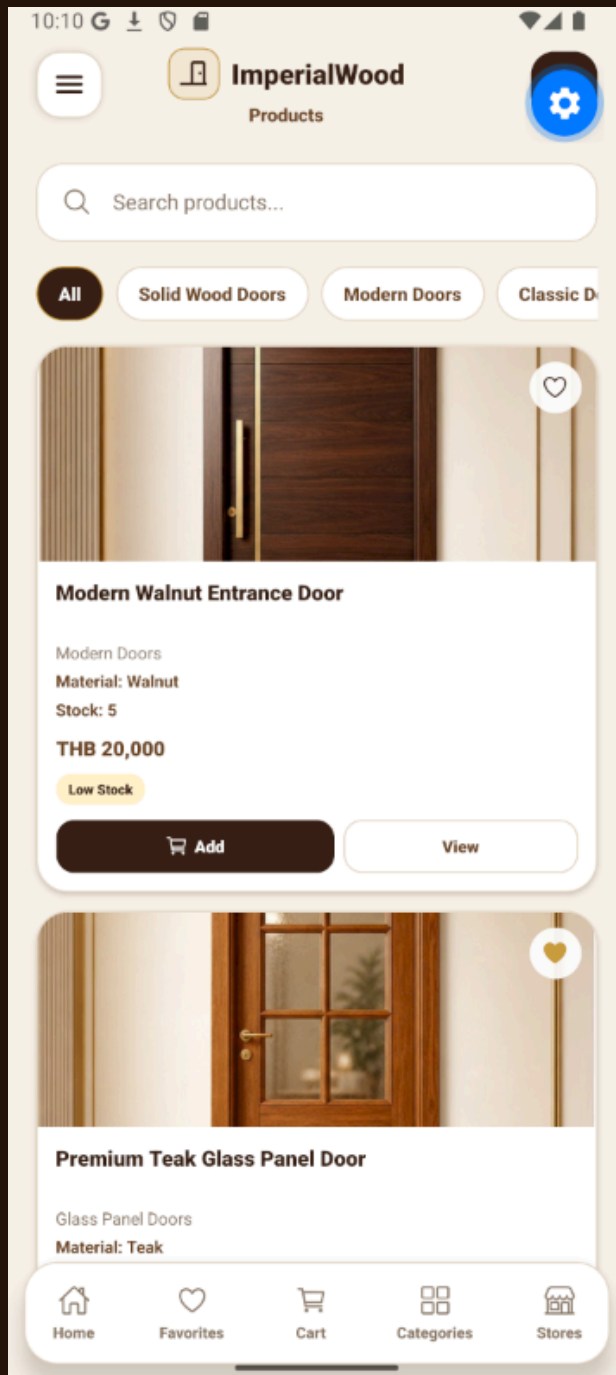
The Register page allows new customers to create an ImperialWood Client account by providing the required personal and login information.

Features & Functions

- Full Name — Enter the customer's full name.
- Username — Create a username for signing in.
- Email — Enter the customer's email address.
- Phone Number — Enter the customer's contact number.
- Password — Create a password for the account.
- Confirm Password — Re-enter the password for confirmation.
- Show/Hide Password — Display or hide the entered password.
- Create Account — Submit the registration information and create a new Client account.

How to Use

Enter Personal Information → Create Username & Password → Confirm Password → Tap Create Account → Account Created



Page-by-Page Description

Products Page

The Products page allows clients to browse, search, and filter the available ImperialWood door products.

Features & Functions

- Search Products — Search for products using keywords.
- Category Filter — Filter products by door category.
- Product Information — Displays the product name, category, material, stock, and price.
- Stock Status — Shows the current availability of each product.
- Favorites — Save preferred products by tapping the heart icon.
- Add — Add a selected product directly to the shopping cart.
- View — Open the selected product to view more detailed information.

How to Use

Search / Select Category → Browse Products → Add to Favorites, Add to Cart, or View Details

Page-by-Page Description

Product Detail Page

The Product Detail page provides detailed information about a selected ImperialWood product, helping clients review its specifications, price, stock, and availability.

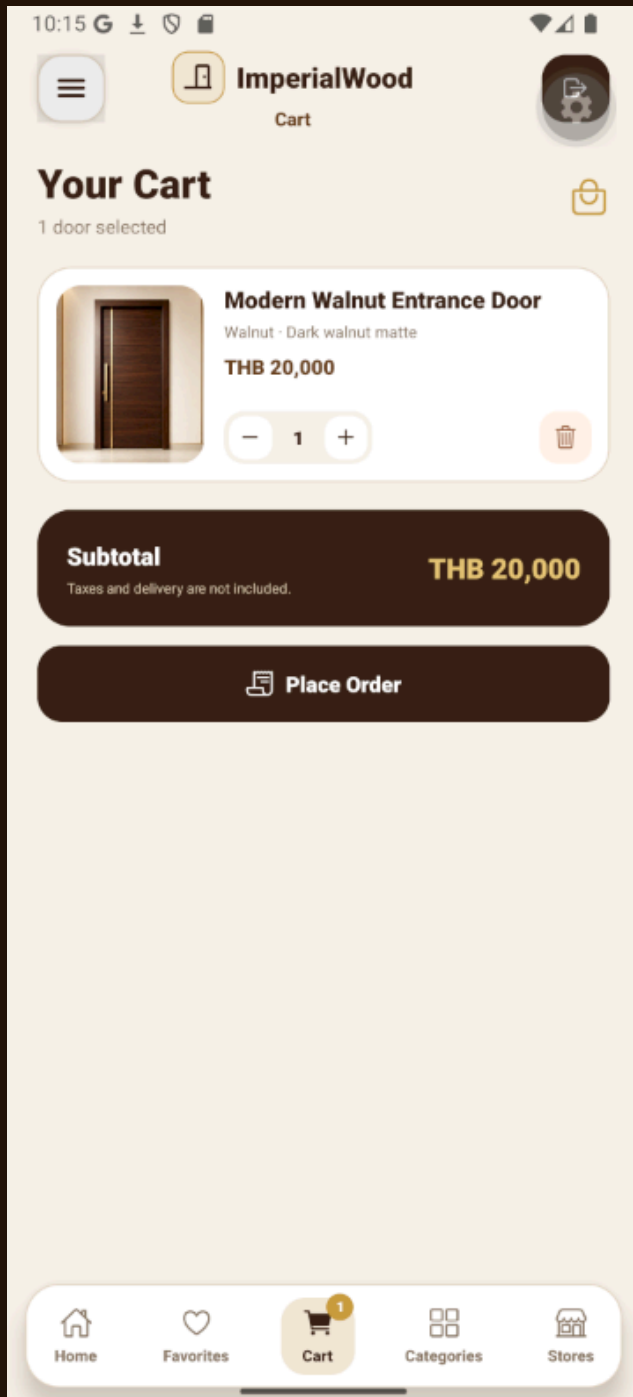
Features & Functions

- Product Image — Displays a large preview of the selected product.
- Product Name & Category — Shows the product name and its category.
- Material & Size — Displays the product material and dimensions.
- Finish — Shows the product's surface finish.
- Price — Displays the current product price.
- Item Code — Shows the unique product identification code.
- Stock — Displays the current quantity available.
- Store Availability — Indicates store availability for the product.
- Description — Provides additional information about the product.
- Status — Displays the current stock status, such as Low Stock.

How to Use

Select Product → Tap View → Review Product Details → Continue Browsing or Use the Navigation Bar





Page-by-Page Description

Cart Page

The Cart page allows clients to review and manage selected products before placing an order.

Features & Functions

- Selected Products — Displays products currently added to the cart.
- Product Information — Shows the product image, name, material, finish, and price.
- Quantity Control — Use the – and + buttons to decrease or increase the quantity.
- Remove Product — Remove an unwanted product from the cart using the delete button.
- Subtotal — Displays the total price of the selected products.
- Place Order — Proceeds with the selected products to create an order.

How to Use

Review Cart → Adjust Quantity → Check Subtotal → Tap Place Order

Page-by-Page Description

Orders Page

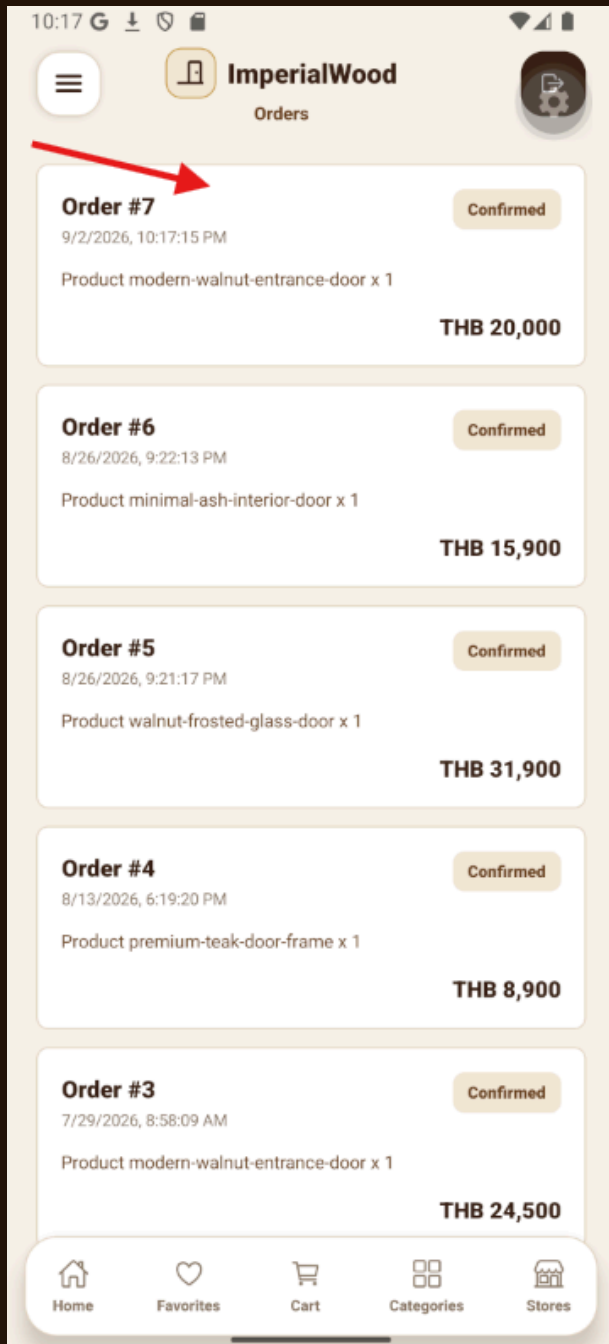
The Orders page allows clients to view their order history and check the status and details of previously placed orders.

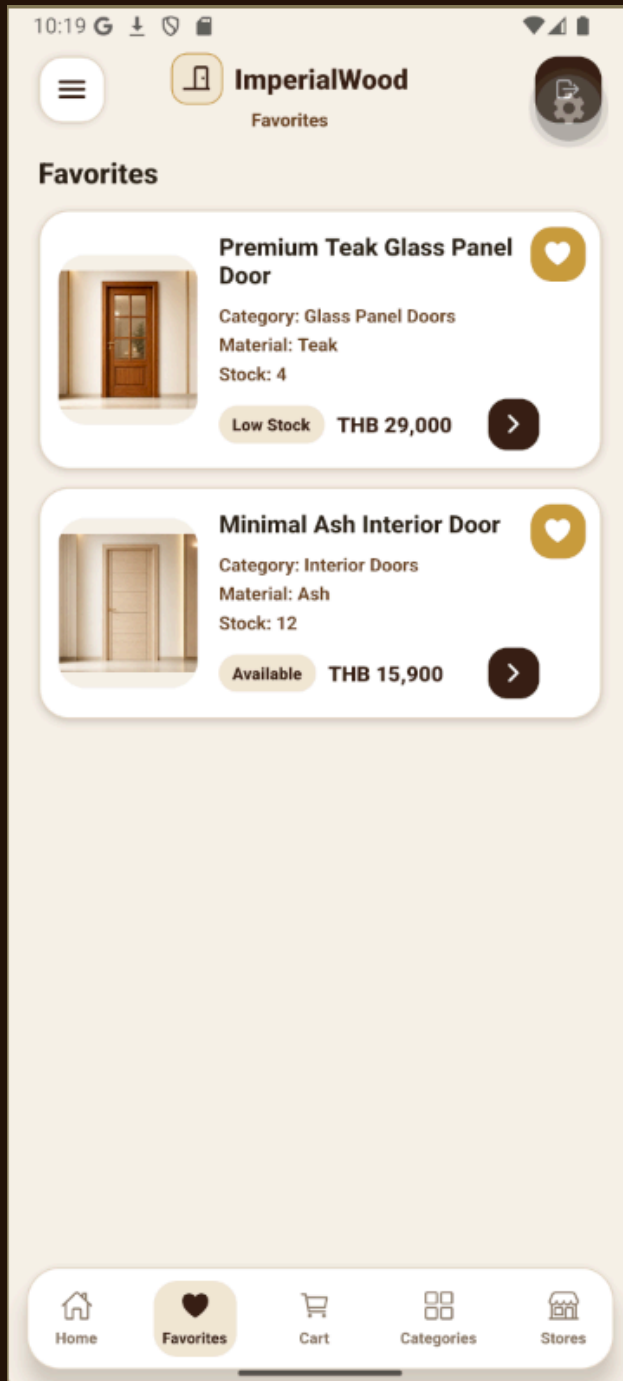
Features & Functions

- Order Number — Displays a unique number for each order.
- Order Date & Time — Shows when the order was placed.
- Product Information — Displays the ordered product and quantity.
- Order Status — Shows the current order status, such as Confirmed.
- Total Price — Displays the total amount of each order.
- Order History — Allows clients to review previous orders.

Order Flow

Add Product → Cart → Place Order → Order Created → Confirmed → View in Orders





Page-by-Page Description

Favorites Page

The Favorites page allows clients to view and manage products they have saved for quick access.

Features & Functions

- Saved Products — Displays products added to the client's favorites.
- Product Information — Shows the product name, category, material, stock, and price.
- Stock Status — Displays product availability, such as Available or Low Stock.
- Remove Favorite — Tap the heart icon to remove a product from Favorites.
- View Product — Tap the arrow button to open the selected product and view its details.

How to Use

Select Favorite → View Saved Products → Check Product Information → Open Product or Remove from Favorites

Admin

Page-by-Page Description

Admin Manage Products Page

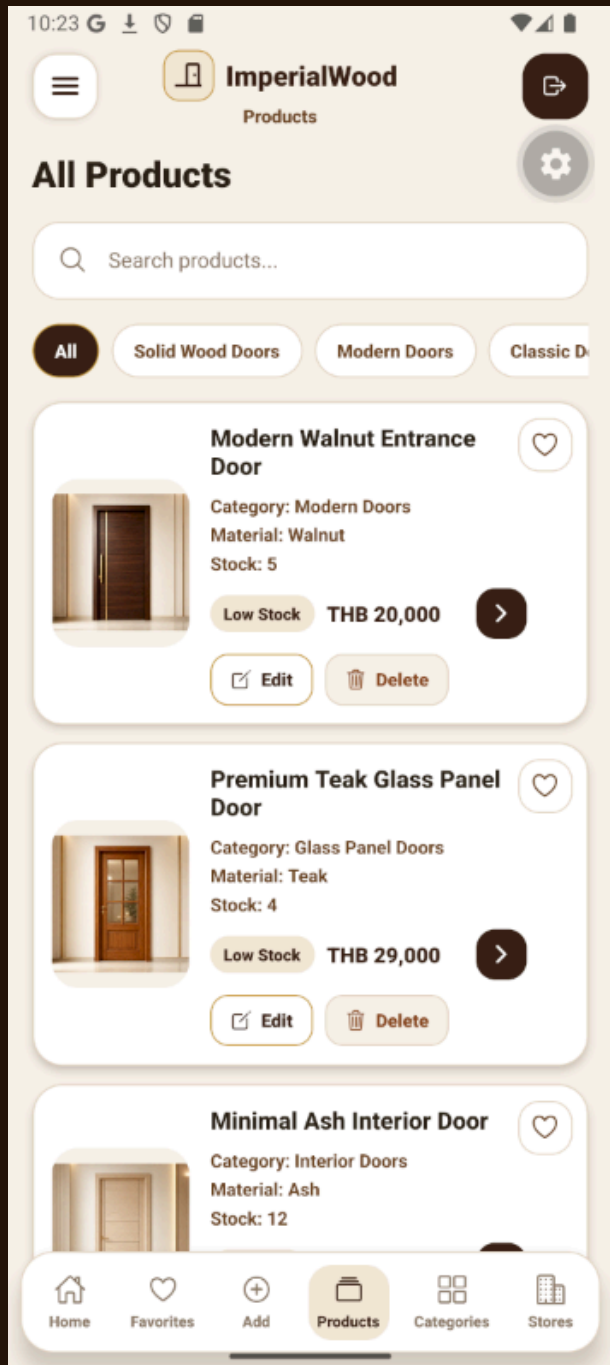
The Admin Products page allows administrators to view, search, filter, and manage all products in the ImperialWood system.

Features & Functions

- Search Products — Search for products using keywords.
- Category Filter — Filter products by product category.
- Product Information — Displays the product name, category, material, stock, price, and stock status.
- View Product — Open the selected product to view detailed information.
- Edit Product — Modify the information of an existing product.
- Delete Product — Remove a product from the system.
- Favorites — Add or remove products from the favorites list.

Admin Workflow

View Products → Search / Filter → Select Product → View, Edit, or Delete



10:25

ImperialWood

Edit Door

Edit Product

Product Name

Modern Walnut Entrance Door

Item Code

IW-002

Category

Accessories Classic Doors Door Frames

Entrance Doors Glass Panel Doors

Interior Doors **Modern Doors**

Solid Wood Doors

Price

20000

Stock Quantity

5

Material

Walnut

Size

90 x 200 cm

Home Favorites Add Products Categories Stores

Page-by-Page Description

Edit Product Page

The Edit Product page allows administrators to modify the information of an existing ImperialWood product.

Features & Functions

- Product Name — Edit the product name.
- Item Code — Modify the unique product code.
- Category — Select the appropriate product category.
- Price — Update the product price.
- Stock Quantity — Update the available stock quantity.
- Material — Modify the product material.
- Size — Update the product dimensions.
- Additional Information — Other product information can be updated by scrolling through the form.
- Save Changes — Save the updated product information to the system.

How to Use

Select Product → Tap Edit → Modify Product Information → Save Changes

Page-by-Page Description

Add Product Page

The Add Product page allows administrators to create and add a new wooden door product to the ImperialWood system.

Features & Functions

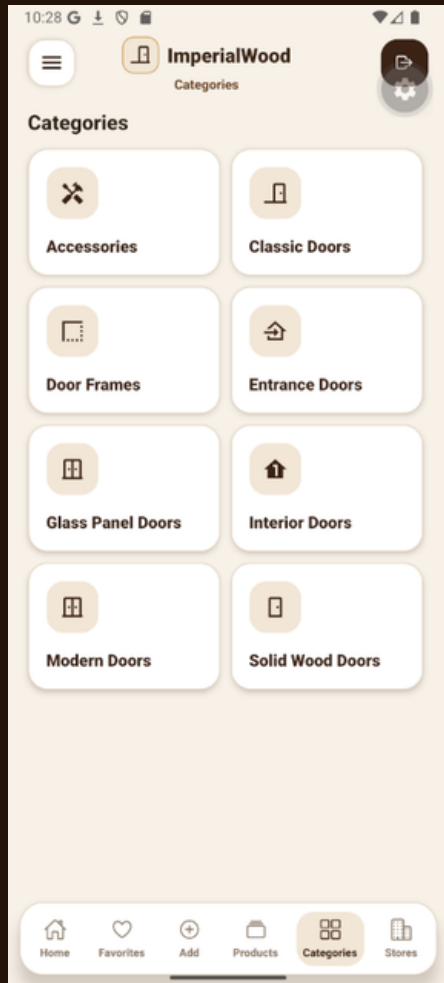
- Product Name — Enter the name of the new product.
- Description — Enter a description of the product.
- Door Category — Select the appropriate product category.
- Item Code — Displays the product item code.
- Price — Enter the product price in THB.
- Stock Quantity — Enter the available quantity of the product.
- Additional Product Information — Additional details can be entered by scrolling through the form.
- Create Product — Submit the information to add the new product to the system.

How to Use

Open Add Product → Enter Product Information → Select Category → Enter Price & Stock → Complete Additional Fields → Create Product

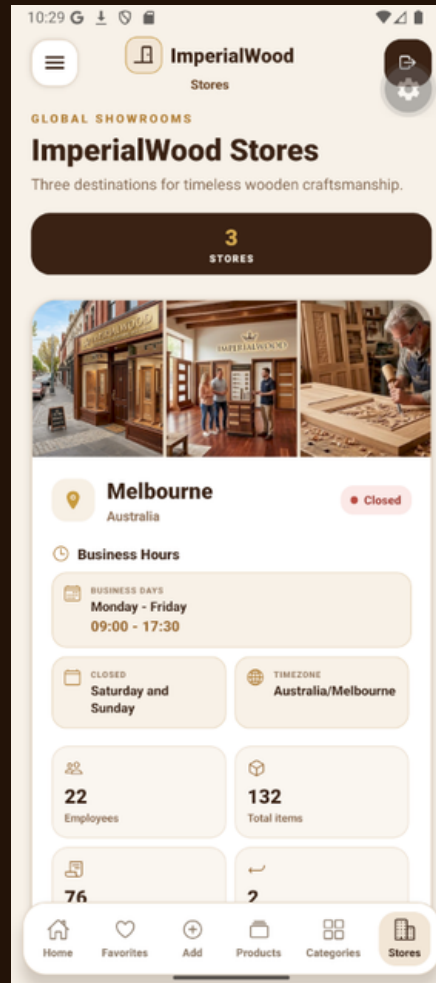
The screenshot shows the 'Add Door' form in the ImperialWood app. At the top, there's a header with the app name 'ImperialWood' and a sub-header 'Add Door'. Below this, the form is divided into several sections: 'Product Name' with a text input field; 'Description' with a larger text input field; 'Door Category' with a grid of buttons including 'Accessories', 'Classic Doors', 'Door Frames', 'Entrance Doors', 'Glass Panel Doors', 'Interior Doors', 'Modern Doors', and 'Solid Wood Doors'; 'Item Code' with a text input field showing 'IW-005'; 'Price' with a text input field and a note 'Price will be shown as THB.'; and 'Stock Quantity' with a text input field. At the bottom, there's a navigation bar with icons for 'Home', 'Favorites', 'Add' (highlighted), 'Products', 'Categories', and 'Stores'.

Additional Application Pages



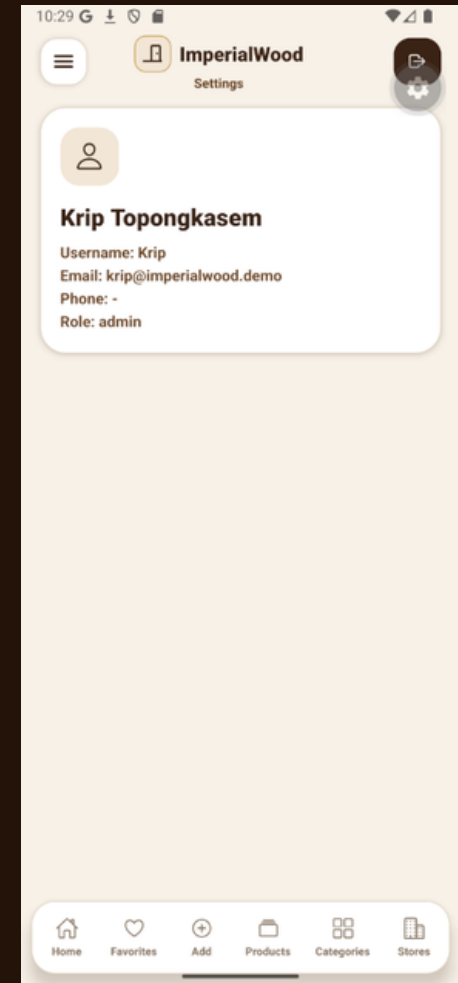
Categories

Allows users to browse products by category, including Accessories, Classic Doors, Door Frames, Entrance Doors, Glass Panel Doors, Interior Doors, Modern Doors, and Solid Wood Doors.



Stores

Displays ImperialWood showroom information, including location, business hours, closed days, time zone, store status, and other store information.



Profile / Settings

Displays the currently logged-in user's account information, including name, username, email, phone number, and user role.



Prepared by
Krip Topongkasem

My Github

